

1.

1.1.

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1.2.

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12 cm^2 g

(1) = +

(2)

(3)

$$
\begin{aligned}
(\ ,\ ) = {} & (11,23),\quad (23,49),\quad (5,10),\quad (50,115),\quad (32,64),\quad (21,45) \\
& (16,35),\quad (39,84),\quad (25,50),\quad (45,100),\quad (5,11),\quad (34,70)
\end{aligned}
$$

2.

2.1.

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(1)

(1)

(2) (2) (2)

(1) (2)

(3) 1 6 (2)

1 2 3 4 5 6 (2) $\Omega = \{1, 2, 3, 4, 5, 6\}$ (3)

(2) Ω (1)

(3) (2)

2.2.

2022 2022 2022 pp. 78, 80, 83

(1)

(2)

(3)

(4) 3

2.3.

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1

$(\ = \ ) = \ \times \ (\ = 1, 2, 3, 4, 5, 6)$ $\times$

(1)

(2) $= 1, 2, 3, 4, 5, 6$ $(\ = \ )$

(3) $[\ ]$

(4) $(\ \geq 4)$

2.4.

2023 pp. 81, 86

$$(\ ) = \begin{cases} \frac{}{\ ^{3}}(\quad \geq 10) \\ 0 \quad (\quad < 10) \end{cases}$$

(1)

(2) $(\ )$

(3) $(\quad \geq \quad) = \frac{1}{2}$

2.5.

2024 pp. 81, 86

$$(\ ) = \begin{cases} (1 - \quad)(0 \leq \quad \leq 1) \\ 0 \quad \text{(otherwise)} \end{cases}$$

(1)

(2) $(\frac{1}{2} < \quad \leq \frac{2}{3})$

(3) $(\ )$

(4) $(\frac{1}{2} \leq \quad < 3)$

2.6.

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(1)

(2)

(3)

(4)

2.7.

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$_1$, $_2$, ⋯, 2

(1) $_1$, $_2$, ⋯,

(2) 2 /

(3)

(4)

(5)

(6) 1

2.8.

2025

(1) 2

(2) 200 60 6 42
78

3.

3.1.

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6 4 1 2

(1) 2

(2) 2

(1) $\frac{6}{10} \quad \frac{5}{9} = \frac{1}{3}$

(2) $\frac{6}{10} \quad \frac{5}{9} + \frac{4}{10} \quad \frac{6}{9} = \frac{3}{5}$

3.2.

2026 p. 107

3 5

(1) 1 1

(a) 2

(b) 2

(c) 2 1

(2) 2

1

(3) (1) (2)

(1) (a) $\frac{3}{8} \quad \frac{2}{7} = \frac{6}{56} = \frac{3}{28}$

(b) $\frac{3}{8} \quad \frac{2}{7} + \frac{5}{8} \quad \frac{3}{7} = \frac{21}{56} = \frac{3}{8}$

(c) $\frac{6/56}{21/56} = \frac{6}{21} = \frac{2}{7}$

(2) $\frac{3}{8}$

(3) 1

3.3.

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1 2 3 4 2

$_1$ $_2$ $= |\ _1 - \ _2|$

(1)

(2)

(3)

(1)

	0	1	2	3	
	$^1\!/_4$	$^3\!/_8$	$^1\!/_4$	$^1\!/_8$	1

(2) $[\ ] = \frac{1}{4}\ 0 + \frac{3}{8}\ 1 + \frac{1}{4}\ 2 + \frac{1}{8}\ 3 = \frac{5}{4}$

(3) $\mathrm{Var}[\ ] = \ [\ ^2] - (\ [\ ])^2 = (\frac{1}{4}\ 0^2 + \frac{3}{8}\ 1^2 + \frac{1}{4}\ 2^2 + \frac{1}{8}\ 3^2) - (\frac{5}{4})^2 = \frac{5}{2} - \frac{25}{16} = \frac{15}{16}$

3.4.

2023 2025 pp. 113 - 114

$(\ ,\ )$

$$(\ ,\ ) = \begin{cases} 6\ ^{-2}\ ^{-3} & (\ \geq 0,\ \ \geq 0) \\ 0 & (\text{otherwise}) \end{cases}$$

≥ 0 ≥ 0

$$_1(\ ) = \int_{-\infty}^{\infty} (\ ,\ )\ \ = \int_0^{\infty} 6\ ^{-2}\ ^{-3}\ \ = 6\ ^{-2}\left[-\frac{1}{3}\ ^{-3}\right]_0^{\infty} = 2\ ^{-2}$$

$$_2(\ ) = \int_{-\infty}^{\infty} (\ ,\ )\ \ = \int_0^{\infty} 6\ ^{-2}\ ^{-3}\ \ = 6\ ^{-3}\left[-\frac{1}{2}\ ^{-2}\right]_0^{\infty} = 3\ ^{-3}$$

$$_1(\ ) = \begin{cases} 2\ ^{-2} & (\ \geq 0) \\ 0 & (\ < 0) \end{cases} \qquad _2(\ ) = \begin{cases} 3\ ^{-3} & (\ \geq 0) \\ 0 & (\ < 0) \end{cases}$$

≥ 0 ≥ 0

$$(\ ,\ ) = \ _1(\ )\ _2(\ )$$

3.5.

2026

$$\int_0^{\infty} \; ^{-} \quad = 1, \qquad \int_0^{\infty} \; ^{-} \quad = 1$$

(1) $(\; , \;) = \{ \; ^{- \; -} \quad (\quad \geq 0, \quad \geq 0)$; 0 (otherwise)

(a) $(\;) \qquad (\;)$

(b)

(2) $(\; , \;) = \{ \; (\quad + \quad) \; ^{- \; -} \quad (\quad \geq 0, \quad \geq 0)$; 0 (otherwise)

(a)

(b)

---

(1) (a) $\geq 0 \qquad \geq 0$

$$(\;) = \int_{-\infty}^{\infty} (\; , \;) \quad = \quad ^{-} \int_0^{\infty} \underbrace{\; ^{-} \qquad}_{=1} \quad = \quad ^{-}$$

$$(\;) = \int_{-\infty}^{\infty} (\; , \;) \quad = \quad ^{-} \int_0^{\infty} \underbrace{\; ^{-} \qquad}_{=1} \quad = \quad ^{-}$$

$(\;) = \{ \; ^{-} \quad (\quad \geq 0)$; 0 (otherwise)’ $\qquad (\;) = \{ \; ^{-} \quad (\quad \geq 0)$; 0 (otherwise)

(b) $(\; , \;) = \quad ^{-} \quad ^{-} \quad = \quad (\;) \quad (\;)$

(2) (a) $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (\; , \;) \qquad = 1$

$$\int_0^{\infty} \int_0^{\infty} (\quad + \quad) \; ^{- \; -} \qquad = \int_0^{\infty} \underbrace{\; ^{-} \qquad}_{=1} \int_0^{\infty} \underbrace{\; ^{-} \qquad}_{=1} + \int_0^{\infty} \underbrace{\; ^{-} \qquad}_{=1} \int_0^{\infty} \underbrace{\; ^{-} \qquad}_{=1} = 2$$

$$= \frac{1}{2}$$

(b) $\geq 0, \quad \geq 0$

$$(\;) = \int_{-\infty}^{\infty} (\; , \;) \quad = \frac{1}{2} \; ^{-} \int_0^{\infty} \underbrace{\; ^{-} \qquad}_{=1} + \frac{1}{2} \; ^{-} \int_0^{\infty} \underbrace{\; ^{-} \qquad}_{=1} = \frac{1}{2}(\quad + 1) \; ^{-}$$

$$(\ ) = \int_{-\infty}^{\infty} (\ , \ ) \quad = \frac{1}{2} - \int_{0}^{\infty} - \ \underset{=1}{\text{sss t}} + \frac{1}{2} - \int_{0}^{\infty} - \ \underset{=1}{\text{ss t}} = \frac{1}{2}(\ + 1) -$$

$= \quad = 0 \qquad (\ , \ ) \neq \ (\ ) \ (\ )$

3.6.

2022 pp. 103 - 104, 113

1 4 2 2 1

$= 0$ $= 1$ 2

$= 0$ $= 1$

(1) $(\ , \ )$

(2)

(1)

	0	1	
0	2⁄5	2⁄15	2⁄3
1	4⁄15	1⁄15	1⁄3
	2⁄3	1⁄3	1

(2) $(\ = 0, \ = 0) \neq (\ = 0) \ (\ = 0)$

3.7.

2022 pp. 67, 114 - 115

2

$$(\ \cap \ ) = (\ ) \ (\ )$$

$(\ \cap \ ) = 0$ 2

2

3.8.

2022 2023 2026 p. 115

3 $\frac{1}{2}$

(1) 1 2

(2) 1 2 2 3

(3) $(\mid)$

(1) $\Omega = (\ , \ )^3$ $(\ , \ , \ )$ 1 2 3 $\equiv$ Ω $\equiv$ $\Omega_0 = [(\ , \ , \ )]_\equiv$ $\Omega_1 = [(\ , \ , \ )]_\equiv$ $\Omega_2 = [(\ , \ , \ )]_\equiv$ $\Omega_3 = [(\ , \ , \ )]_\equiv$ $|\Omega_0| = \binom{3}{0} = 1$ $|\Omega_1| = \binom{3}{1} = 3$ $|\Omega_2| = \binom{3}{2} = 3$ $|\Omega_3| = \binom{3}{3} = 1$ $|\Omega| = 8$ $\frac{4/8}{7/8} = \frac{4}{7}$

(2) $= \{(\ , \ , \ ), (\ , \ , \ )\}$ $= \{(\ , \ , \ ), (\ , \ , \ )\}$ $\cap$ $\cap$ $= \{(\ , \ , \ )\}$ $|\Omega|$ $(\ ) = \frac{1}{4}$ $(\ ) = \frac{1}{4}$ $(\ \cap \ ) = \frac{1}{8}$ $(\ \cap \ ) = (\ ) (\ )$

(3) $$(\ \mid \ ) = \frac{(\ \cap \ )}{(\ )} = \frac{1/8}{1/4} = \frac{1}{2}$$

3.9.

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2 4 2 7 2 9

(1)

(2)

2 $= 4$ $+ \ = 7$ $+ \ = 9$

(1) $\bullet$ $(\ ) = \frac{1}{6}$

- $(\ ) = (\ + \ = 7) = \frac{1}{6}$
- $(\ \cap\ ) = (\ = 4,\ + \ = 7) = (\ = 4,\ = 3) = \frac{1}{36}$

  $(\ \cap\ ) = (\ )(\ )$

(2)

- $(\ ) = \frac{1}{6}$
- $(\ ) = (\ + \ = 9) = \frac{1}{9}$
- $(\ \cap\ ) = (\ = 4,\ + \ = 9) = (\ = 4,\ = 5) = \frac{1}{36}$

  $(\ \cap\ ) \neq (\ )(\ )$

3.10.

2023 pp. 72 - 73, 109 - 110, 114 - 115

2 $(\ ) =$ $(\ ) =$

(1) $(\quad)$

(2) $(\ \cap \ |\quad)$

(1) $(\quad) = \ + \ -$

(2) $(\ \cap \ |\quad) = \frac{(\ \cap\ )}{(\quad)} = \frac{-}{+\ -}$

3.11.

2025

1 5

$= \ -$

(1)

(2)

(3)

(4)

3.12.

2025 pp. 118 - 119

(1) 2

(2) 500 1

2 $_1$ $_2$ 3 : 1

$_1$, $_2$ 0.98 0.90

0.01

$_1$

3.13.

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3 5 1

2

(1) 1

(2) 2 1

3.14.

2023 pp. 118 - 119

1 10% 3

(1)

(2) 2

(1) $\frac{9}{10}\ \frac{9}{10}\ \frac{9}{10} = 0.7290$

(2) $\dfrac{\frac{9}{10}\ \frac{1}{10}}{\frac{1}{10} + \frac{9}{10}\ \frac{1}{10} + \frac{9}{10}\ \frac{9}{10}\ \frac{1}{10}} = \dfrac{90}{271} \approx 0.3321$

3.15.

2022 pp. 118 - 119

5 4 3

1 100 g 45 g 30 g

(1) 1

40 g

(2) 2

100 g 2

(3) 3

150 g 3

(1) $\dfrac{\binom{5}{1}}{\binom{12}{1}-\binom{3}{1}} = \dfrac{5}{9}$

(2) $\dfrac{\binom{5}{2}}{\binom{12}{2}-\binom{7}{2}} = \dfrac{10}{66-21} = \dfrac{2}{9}$

(3) $\dfrac{\binom{5}{3}}{\binom{12}{3}-\binom{7}{3}} = \dfrac{10}{220-35} = \dfrac{10}{185} = \dfrac{2}{37}$

3.16.

2022 2024 pp. 118 - 119

A 0.1%

A /

99% 10%

(1)

(2) A 1

A 2

(1) $(\ |\ ) = \dfrac{(\ )\ (\ |\ )}{(\ )\ (\ |\ ) + \ (\ )\ (\ |\ )} = \dfrac{0.001\ \ 0.99}{0.001\ \ 0.99 + 0.999\ \ 0.10} = \dfrac{99}{10089} \approx 0.0098$

(2) $(\ |\ ) = \dfrac{(\ )\ (\ |\ )}{(\ )\ (\ |\ ) + \ (\ )\ (\ |\ )} = \dfrac{0.001\ \ 0.99^2}{0.001\ \ 0.99^2 + 0.999\ \ 0.10^2} = \dfrac{9801}{109701} \approx 0.0893$

3.17.

2023 pp. 118 - 119

99%

99% 0.5%

1

(1)

(2)

(3)

(4)

(1) $(\cap) = ( ) (|) = 0.005 \quad 0.99 = 0.00495$

(2) $(\cap) = ( ) (|) = 0.995 \quad 0.01 = 0.00995$

(3) $( ) = (\cap) + (\cap) = 0.00495 + 0.00995 = 0.0149$

$$(|) = \frac{(\cap)}{( )} = \frac{0.00995}{0.0149} \approx 0.6678$$

(4) $(|) = \frac{(\cap)}{( )} = \frac{( ) (|)}{1 - ( )} = \frac{0.005 \quad 0.01}{0.9851} \approx 0.0001$

3.18.

2023 pp. 118 - 119

S S

K

- S K 99% 1%
- S K 99% 1%

(1) K S

(2) (1)

(1)

$$(|) = \frac{( ) (|)}{( ) (|) + ( ) (|)} = \frac{\frac{1}{10000} \quad \frac{99}{100}}{\frac{1}{10000} \quad \frac{99}{100} + \frac{9999}{10000} \quad \frac{1}{100}} = \frac{99}{10098} \approx 0.0098$$

(2)

3.19.

2025 pp. 118 - 119

A

B

(1) $(\ |\ ) = 0.98$ $(\ |\ \ ) = 0.005$ $(\ \ ) = 0.001$

(2) $(\ |\ \ ) = 0.003$ $(\ \ ) = 0.0005$ $(\ |\ ) \geq 0.142$ $(\ |\ )$

(1)

$$(\ |\ ) = \frac{(\ )\ (\ |\ )}{(\ )\ (\ |\ ) + \ (\ \ )\ (\ |\ \ )} = \frac{0.001\ \ 0.98}{0.001\ \ 0.98 + 0.999\ \ 0.005} = \frac{980}{5975} \approx 0.1640$$

(2) (1) $(\ |\ ) = \dfrac{0.0005\ \ \ (\ |\ )}{0.0005\ \ \ (\ |\ ) + 0.9995\ \ 0.003} \geq 0.142$

$$(\ |\ ) \geq \frac{1}{0.0005}\ \ \frac{1}{1 - 0.142} + 0.142\ \ 0.9995\ \ 0.003 \approx 0.9925$$

3.20.

pp. 118 - 119

500 1 XXX 2

$_1$ $_2$ $_1 : \ _2 = 3 : 1$

$\boldsymbol{K_1}$ 0.98

$\boldsymbol{K_2}$ 0.90

0.01

(1) $_1$

(2) (1)

1 2 1 2

(1) $(\ _1) = \frac{1}{500}\ \overset{0}{\frac{3}{4}} = \frac{3}{2000}$ $(\ _2) = \frac{1}{500}\ \frac{1}{4} = \frac{1}{2000}$ $(\ _0) = \frac{499}{500} = \frac{1996}{2000}$

$$(\ ) = \underbrace{\frac{3}{2000}\ \frac{98}{100}}_{(\ _1\cap\ )} + \underbrace{\frac{1}{2000}\ \frac{90}{100}}_{(\ _2\cap\ )} + \underbrace{\frac{1996}{2000}\ \frac{1}{100}}_{(\ _0\cap\ )} = \frac{294+90+1996}{200000} = \frac{2380}{200000}$$

$$(\ _1|\ ) = \frac{(\ _1\cap\ )}{(\ )} = \frac{294}{2380} \approx 0.1235$$

(2)

3.21.

2023 pp. 122 - 124

$0 < \ < 1$

$$(\ ) = \begin{cases} & (\ = 1) \\ 1 - & (\ = -1) \end{cases}$$

$=$

(1) $[\ ]$

(2) $[(\ - [\ ])(\ - [\ ])]$

(3)

(4) (3) $\Pr[\ + \ + \ \le 2]$

(1) $[\ ] = [\ ] = - (1 - \ ) = 2\ - 1$ $[\ ] = [\ ] = [\ ]\ [\ ] = (2\ - 1)^2$

(2) $[\ ] = [\ ^2\ ] = [\ ]$

$$\mathrm{Cov}(\ ,\ ) = [\ ] - [\ ]\ [\ ] = (2\ - 1)[1 - (2\ - 1)^2] = 4\ (1 - \ )(2\ - 1)$$

(3) $\mathrm{Cov}(\ ,\ ) = 0$ $0 < \ < 1$ $= \frac{1}{2}$

$(\ =\ ,\ =\ ) = (\ =\ )\ (\ =\ ) = \frac{1}{4}$

(4) $(\ ,\ ) = (1, 1)$ $+ \ + \ = 3 > 2$

$$\Pr[\ + \ + \ \le 2] = 1 - \Pr[\ + \ + \ > 2] = \frac{3}{4}$$

3.22.

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2

(1)

(2) Cov(,)

(3)

	= 1	= 10	= 100
= 1			
= 2	0		
= 3			

(1) 8 = 1 $= \frac{1}{8}$

(2)

- $[\ \ ] = \frac{3}{8}\ 1 + \frac{2}{8}\ 2 + \frac{3}{8}\ 3 = 2$
- $[\ \ ] = \frac{2}{8}\ 1 + \frac{3}{8}\ 10 + \frac{3}{8}\ 100 = \frac{83}{2}$
- $[\ \ \ \ ] = \frac{664}{8} = 83$

$$\mathrm{Cov}(\ ,\ ) = \ [\ \ ] - \ [\ ]\ [\ ] = 83 - 2\ \frac{83}{2} = 0$$

(3) $(\ = 2) = \frac{1}{4}$ $(\ = 1) = \frac{1}{4}$ $(\ = 2,\ = 1) = 0$

$$(\ = 2,\ = 1) \neq \ (\ = 2)\ (\ = 1)$$

3.23.

2023 p. 127

1 $(\ + 1)$

$$(\ _1\ )\ \underbrace{(\frac{1}{6})}_{2\times 1}\ \underbrace{(\frac{1}{6})^{-1}_{s\ \ t}}_{1\times(\ -1)} = \frac{}{6}$$

4.

4.1.

2025

2 (2,)

$$(2,\ ) \qquad (\ =\ ) = \binom{2}{\ } \quad (1-\ )^{2-}$$

$$[\ ] = \sum_{=0}^{2} \quad (\ =\ ) = \quad (\ =1) + 2 \quad (\ =2)$$

$$= \binom{2}{1} \quad (1-\ ) + 2 \quad \binom{2}{2} \quad ^2$$

$$= \quad [2(1-\ ) + 2\ ] = 2$$

$$\mathrm{Var}(\ ) = \quad [(\ -\ [\ ])^2] = \quad [(\ -2\ )^2]$$

$$= \sum_{=0}^{2} (\ -2\ )^2 \quad (\ =\ )$$

$$= (0-2\ )^2 \quad (\ =0) + (1-2\ )^2 \quad (\ =1) + (2-2\ )^2 \quad (\ =2)$$

$$= 4\ ^2 \binom{2}{0} (1-\ )^2 + (1-2\ )^2 \binom{2}{1} \quad (1-\ ) + 4(1-\ )^2 \binom{2}{2} \quad ^2$$

$$= 8\ ^2(1-\ )^2 + 2\ (1-\ )(1-2\ )^2 = 2\ (1-\ )$$

4.2.

2024 p. 141

1 6

(1) 10 5

(2) 100 25

(1) $(10, \frac{1}{6})$

$$(\ \geq 5) = \sum_{=5}^{10} \quad (\ =\ ) = \sum_{=5}^{10} \binom{10}{\ } \left(\frac{1}{6}\right) \quad \left(\frac{5}{6}\right)^{10-} \quad = \frac{1}{6^{10}} \sum_{'=0}^{5} \binom{10}{'} 5^{'}$$

$$= \frac{1}{6^{10}} (1 \quad 1 + 10 \quad 5 + 45 \quad 25 + 120 \quad 125 + 210 \quad 625 + 252 \quad 3125)$$

$$= \frac{934926}{60466176} = 0.01546\ldots \approx 0.0155$$

(2) $(100, \frac{1}{6})$ $[\ ] = \frac{50}{3}$ $[\ ] = \frac{5\sqrt{5}}{3}$

$(1 - \ ) >$ $= \frac{50}{3} > 5$

$(100, (5\sqrt{5}/3)^2)$

$$(\ \geq 25) \approx (\ > 24.5) = (\ > \frac{24.5 - 50/3}{5\sqrt{5}/3}) = (\ > \frac{23.5}{5\sqrt{5}})$$

$$= (\ > 0.94\sqrt{5}) \approx (\ > 2.102) \approx 1 - 0.9821 = 0.0179$$

4.3.

2023 2026 pp. 93 - 95, 141

2 $\mathrm{Bin}(\ ,\ )$ $= (\text{—})$

(1) $(|\ -\ | \leq 0.02) \geq 0.95$

(2) (1) (2)

(3)

(3) (1) -

(4) (2) (3)

(1) $|\ -\ | \leq 0.02$ 0.02

$(|\ -\ | \leq 0.02) \geq 0.95$ 0.02

0.95

(2) $\mathrm{Var}(\ ) = \frac{(1 - \ )}{}$ $= \frac{0.02}{\sqrt{\mathrm{Var}(\ )}}$

$$(|\ -\ | \leq 0.02) = (|\ -\ | < \sqrt{\mathrm{Var}(\ )}) \geq 1 - \frac{1}{^2} = 1 - \frac{(1 - \ )}{0.02^2} \geq 0.95$$

$\frac{(1 - \ )}{0.02^2} \leq 0.05$ $\geq 50,000\ (1 - \ )$

$0^2 < (1 - \ ) \leq (\frac{1}{2})^2$ $\geq 12,500$

(3)

$(\ , \mathrm{Var}(\ ))$

$$(| \ - \ | \le 0.02) = \ (| \frac{-}{\sqrt{\mathrm{Var}(\ )}} | \le \frac{0.02}{\sqrt{\mathrm{Var}(\ )}}) = \ (| \ | \le \frac{0.02}{\sqrt{\mathrm{Var}(\ )}}) \ge 0.95$$

$$(| \ | \le 1.96) \approx 0.95$$

$$\frac{0.02}{\sqrt{\mathrm{Var}(\ )}} \ge 1.96 \qquad \ge \left(\frac{1.96}{0.02}\right)^2 (1 - \ ) = 9,604 \ (1 - \ ) \ge 2,401$$

(4)

$$(| \ - \ | \le \ ) \ge 1 -$$

$$= \sqrt{\frac{(1 - \ )}{}}$$

$$(| \ - \ | \le \ ) = \ (| \ - \ | < - \ ) \ge 1 - (-)^2$$

$$(| \ - \ | \le \ ) \ge 1 - (-)^2 \ge 1 -$$

$$\ge (-)^2 = \frac{(1 - \ )}{2} \qquad \ge \frac{(1 - \ )}{2} \ge \frac{1}{4 \ 2}$$

$$(| \ - \ | \le \ ) = \ (| \frac{-}{} | \le -) = \ (| \ | \le -) = 2\Phi(-) - 1$$

$$(| \ - \ | \le \ ) \ge 2\Phi(-) - 1 \ge 1 -$$

$$\Phi\left(\ \sqrt{\frac{}{(1 - \ )}} \right) \ge 1 - \frac{}{2}$$

$$(\ ) \qquad 1 - \ = \Phi(\ (\ ))$$

$$\sqrt{\frac{}{(1 - \ )}} \ge \ \left(\frac{}{2}\right) \qquad \ge \frac{(1 - \ )}{2} \left[\ \left(\frac{}{2}\right) \right]^2 \ge \frac{1}{4 \ 2} \left[\ \left(\frac{}{2}\right) \right]^2$$

4.4.

2024 pp. 143 - 145, 153

(1) :

$$(\ ) = \frac{1}{\sqrt{2}}\ ^{-\frac{(\ -\ )^2}{2\ ^2}} \quad (-\infty < \quad < \infty)$$

$= (\ )$ $(\ ,\ )$

(2) /

(3)

(1) $' = -\frac{(\ -\ )}{2}$ $'' = \frac{(\ -\ )-}{}\ \frac{(\ -\ )+}{}$

(2) $(\ )$

(3)

0.5

4.5.

2022 pp. 121 - 122, 143

(1) $(170, 6^2)$

(a) $= \frac{+}{2}$

(b)

(2)

$$(\ ) = \frac{1}{2} \quad (0 \leq \quad \leq 2\ ), \qquad (\ ) = \frac{1}{12} \quad (-3 \leq \quad \leq 9)$$

$=$

(1) (a) $[\ ] = [\frac{1}{2}\ + \frac{1}{2}\ ] = \frac{1}{2}\ [\ ] + \frac{1}{2}\ [\ ] = \frac{1}{2}\ 170 + \frac{1}{2}\ 170 = 170$

(b) $\mathrm{Var}(\ ) = \mathrm{Var}(\frac{1}{2}\ + \frac{1}{2}\ ) = (\frac{1}{2})^2 \mathrm{Var}(\ ) + (\frac{1}{2})^2 \mathrm{Var}(\ ) = \frac{1}{4}\ 6^2 + \frac{1}{4}\ 6^2 = 18$

(2) $[\ ] = \int_0^2\ (\ )\ = \frac{1}{2}\int_0^2\ = \frac{1}{4}[\ ^2]_0^2 = \frac{(2\ )^2}{2} =$

$$[\ ] = \int_{-3}^{9}\ (\ )\ = \frac{1}{12}\int_{-3}^{9}\ = \frac{1}{24}[\ ^2]_{-3}^{9} = \frac{1}{24}(9^2 - (-3)^2) = 3$$

$$[\quad] = \quad [\quad\quad] = \quad 3 = 3$$

4.6.

2024 pp. 147 - 151

(1) $(\ ,\ ^2)$

(a) $(\ -\ \leq\ \leq\ +\ )$

(b) $(\ -2\ \leq\ \leq\ +2\ )$

(c) $(\ -3\ \leq\ \leq\ +3\ )$

(2) (

)

(3)

(a) (0.01)

(b) $\Phi(\ _0) = 0.01$ $_0$

(c) (0.005)

(1) $(\ +\ \leq\ \leq\ +\ ) = \Phi(\ ) - \Phi(\ )$ $\Phi(\ ) - \Phi(-\ ) = 2\Phi(\ ) - 1$

(a) $(\ -\ \leq\ \leq\ +\ ) = \Phi(1) - \Phi(-1) = 0.6826$

(b) $(\ -2\ \leq\ \leq\ +2\ ) = \Phi(2) - \Phi(-2) = 0.9544$

(c) $(\ -3\ \leq\ \leq\ +3\ ) = \Phi(3) - \Phi(-3) = 0.9974$

(2) 3 $[\ -3\ ,\ +3\ ]$ 3

(3) (a) 2.327

(b) -2.327

(c) 2.575

4.7.

2024 pp. 149 - 151

(1) $(0, 1)$

$$(|\ | \geq\ ) = 2\ (\ \geq\ )$$

(2) $(\ ,\ ^2)$

$$(| \quad - \quad | \geq \frac{}{4})$$

4

(3)

—

$$(|\overline{\ } - \quad | \geq \frac{}{4}) \leq 0.02$$

(1) $(\ ) = \frac{1}{\sqrt{2}} \ ^{-\frac{2}{2}}$ $(- \ ) = (\ )$

$$(| \ | \geq \ ) = \int_{-\infty}^{-} (\ ) + \int^{\infty} (\ ) = \int^{\infty} (\ ) + \int^{\infty} (\ ) = 2 \ (\ \geq \ )$$

(2) $(| \quad - \quad | \geq \frac{}{4}) = (| \ | \geq \frac{1}{4}) \approx 0.8026 \approx 0.803$

(3) $[\overline{\ }] =$ $[\overline{\ }] = \sqrt{}$

$$(|\overline{\ } - \quad | \geq \frac{}{4}) = (| \ | \geq \frac{\sqrt{\ }}{4}) \leq 0.02$$

$$(\ \leq 2.33) = 0.9901 \qquad (| \ | > 2.33) \approx 0.0198 \approx 0.02$$

$$\frac{\sqrt{\ }}{4} \geq 2.33 \qquad \geq (2.33 \ \ 4)^2 = 86.56$$

$= 87$

4.8.

2024 2026 pp. 149 - 150

4

350,198 66.31

23.55 $(66.31, (23.55)^2)$

(1) $(\ \geq 80)$

(2) (3-a)

(3) 80 95

(4) 20%

(1) $= 350,198$ $= 66.31$ $= 23.55$ $(\ , \ ^{2})$

$= \frac{\ -\ }{\ }$ $(0, 1)$ $(\ \geq 80) = (\ \geq$

$\frac{80 -}{\ }) \approx (\ \geq 0.58)$ $(\ = 80) \approx 0$ $(\ \geq$

$80) = (\ \geq 0.58) = 1 - \Phi(0.58)$ $\Phi(0.58) \approx 0.7910$

$(\ \geq 80) \approx 1 - 0.7910 = 0.2090$

(2) 1

80 80

(3) $[80, 90]$ $(80 \leq \ \leq 90)$

$$(80 \leq \ \leq 90) = (\ \leq \frac{90 -}{\ }) - (\ \leq \frac{80 -}{\ })$$

5 $$= (\ \leq \frac{90 - 66.31}{2}$$

$$-\frac{\quad - 120}{10} \approx 1.65 \qquad\qquad \approx 120 - 1.65 \times 10 = 103.5$$

(2) $(100, 15^2)$ $(\quad \geq \quad) = 1 - \Phi\left(\frac{103.5 - 100}{15}\right) \approx 1 - 0.5910 = 0.4090$

4.10.

2022 pp. 153 - 154

$$(42 \leq \quad \leq 78) = \quad (| \quad - 60| < 18) = \quad (| \quad - \quad | < 3 \quad) \geq 1 - \frac{1}{3^2} = \frac{8}{9}$$

(2) =

$$= \quad (42 \leq \quad \leq 78) \geq \quad \times \frac{8}{9} = 177.7\ldots$$

178

(3) $(\; , \; ^2)$

$$(42 \leq \quad \leq 78) = \quad (-3 \leq \quad \leq 3) = 0.9974$$

5.

5.1.

2023 2025 pp. 168 - 169

2

$(,)$ $(=) = ( )$ $(1 -)^{-}$ $(1 -)$
$(, (1 -))$
$( )$ $(1 -)^{-} \to$
$^{-}$ $\frac{}{!}$

5.2.

2024 pp. 93, 168 - 169

1 ATM ATM 1 40
ATM 1 60
ATM
/

60

75

90

105

120

/ /1

$= -$ < 1 $=$
$\frac{}{1 -}$ $=$ $^{-1}$ $= \frac{^{-1}}{1 -}$

$$= \frac{1/90}{1 - 60/90}\ \mathrm{h} = \frac{1}{30}\ \mathrm{h} = 120\ \mathrm{s}$$

5.3.

2023 pp. 166 - 169

(1) 3 5

(2) 1 2 1 4

---

(1) 5 $(5, 0.3)$

$$(\ge 1) = 1 - (= 0) = 1 - (0.7)^5 = 1 - 0.16807 = 0.83193$$

(2) 1 (2)

$$(\ge 4) = 1 - (0 \le \le 3) = 1 - {}^{-2}\left(1 + 2 + 2 + \frac{4}{3}\right) = 1 - \frac{19}{3} {}^{-2}$$

$$\approx 1 - \frac{19}{3} \frac{18}{133} = \frac{57}{399} \approx 0.1429$$

$${}^2 = 1 + \cfrac{2}{(1-1) + \cfrac{1}{3 + \cfrac{1}{5 + \cfrac{1}{7 + \cfrac{1}{9 +}}}}} \approx 1 + \cfrac{2}{0 + \cfrac{1}{3 + \cfrac{1}{5 + \cfrac{1}{7 + 0}}}} = \frac{18}{399} \qquad (5.1)$$

5.4.

2025

4

(1) 1 3

2

(2) $(= 3) = 5 \ (= 5)$

(≤ 3)

---

(1) (3)

$$(\ge 2) = 1 - (0 \le \le 1) = 1 - {}^{-3}(1 + 3) = 1 - \frac{4}{3}$$

$$\approx 1 - 4 \frac{221}{4439} = \frac{3555}{4439} \approx 0.8009$$

$$^{3} \approx 1 + \cfrac{6}{(2-3) + \cfrac{9}{6 + \cfrac{9}{10 + \cfrac{9}{14 + \cfrac{9}{18+0}}}}} = \frac{4439}{221} \tag{5.2}$$

(2) $(\quad = 3) = \quad ^{-} \quad \frac{^{3}}{3!} = \quad ^{-} \quad \frac{^{3}}{6}$ 5 $(\quad = 5) = 5 \quad ^{-} \quad \frac{^{5}}{5!} = \quad ^{-} \quad \frac{^{3}}{6} \left(\frac{}{2} \right)^{2}$ $= 2$

(2)

$$(\quad \leq 3) = \quad ^{-2} \left(1 + 2 + 2 + \frac{4}{3} \right) \approx \frac{18}{133} \quad \frac{19}{3} = \frac{342}{399} \approx 0.8751$$

(5.1)

5.5.

2023 pp. 166 - 169

1% 200

(1)

(2)

(3) 3

(2) (3)

(1) $(\quad = \quad) = (\quad) \left(\frac{1}{100} \right) \left(\frac{99}{100} \right)^{200-} \approx \quad ^{-2} \frac{2}{!}$

(2) $(\quad = 0) \approx \quad ^{-2} \approx \frac{18}{133} \approx 0.1353$

(3) $(\quad \geq 3) = 1 - \quad (0 \leq \quad \leq 2) = 1 - \quad ^{-2}(1 + 2 + 2) \approx 1 - 5 \quad \frac{18}{133} = \frac{43}{133} \approx 0.3233$

5.6.

2024 pp. 153 - 154, 166 - 169

4

(1) 240 6 38 45

(2) 0.1% 500 3

(1) $= 240$ $= \frac{1}{6}$ 6 $(\ , \ )$ $=$ 40 $(1 - \ ) = 200$ 5

$$(\ , \ (1 - \ )) = (40, \frac{100}{3})$$

$$(38 \le \ \le 45) \approx (37.5 \le \ \le 45.5) = (\frac{37.5 - 40}{10/\sqrt{3}} \le \ \le \frac{45.5 - 40}{10/\sqrt{3}})$$
$$= (-0.25\sqrt{3} \le \ \le 0.55\sqrt{3}) \approx (-0.43 \le \ \le 0.95)$$
$$= 0.8289 - 0.3336 = 0.4953$$

(2) $= 500$ $= \frac{1}{1000}$ $(\ , \ )$ $(\ )$

$$= \ = \frac{1}{2}$$

$$(\ \ge 3) = 1 - (0 \le \ \le 2) = 1 - {}^{-\frac{1}{2}}(1 + \frac{1}{2} + \frac{1}{8}) = 1 - {}^{-\frac{1}{2}}\frac{13}{8}$$
$$\approx 1 - \frac{12\ 2^2 - 6\ 2 + 1}{12\ 2^2 + 6\ 2 + 1}\frac{13}{8} = 1 - \frac{37}{61}\frac{13}{8} = \frac{7}{488} \approx 0.0143$$

$$\sqrt{\ } \approx 1 + \frac{2}{(2\ - 1) + \frac{1}{6\ + 0}} = \frac{12\ ^2 + 6\ + 1}{12\ ^2 - 6\ + 1} \tag{5.3}$$

5.7.

2026

$= 0.001$ $= 500$

$${}^{-0.5} \approx 0.6065, \qquad \sqrt{0.5} \approx 0.7071$$

(1) (a)

(b) $[\ ]$ $\mathrm{Var}(\ )$

(2)

(a)

(b) $(\ \ge 3)$

(3)

(a)

(b) $(\quad \geq 3)$

(4)

(a) (2) (3)

(b)

(1) (a) $(500, 0.001)$

(b) $[\quad] = \quad = 0.5$ $\mathrm{Var}(\quad) = \quad (1 - \quad) = 0.4995$

(2) (a)

(5) (4)

(6) $=$ $\to \infty$ (4) $(\ \& \ , \ )$

$(\ ; \ )$ $\lim_{\to \infty} (1 + \text{—})$ $=$

(1) $(1 - \ )$ $(1 - \ ) = {}^3(1 - \ )^2$

(2)

(3) $[\ ] = \sum_{=0}^{-1} (1 - \ ) + = \frac{}{1 -}(1 - \ )$

(4) $(\ \& \ , \ ) = (\ ) \ (1 - \ )^{-}$

(5) $[\ ] = \sum_{=0} (\ \& \ , \ ) =$

(6) $=$

$$\lim_{\to \infty} (\ ) \ (1 - \ )^{-}$$

$$= \lim_{\to \infty} \frac{!}{!(\ - \ )!} (1 - \ )^{-}$$

$$= \lim_{\to \infty} \frac{1}{!} (\ - 1)(\ - 2) \ (\ - \ + 1) (\text{—}) \ (1 - \text{—})^{-}$$

$$= \lim_{\to \infty} \frac{}{!} (1 - \frac{1}{}) (1 - \frac{2}{}) \ (1 - \frac{-1}{}) (1 - \text{—})^{-} \ (1 + \frac{-}{})$$

$$= \frac{}{!}^{-}$$

6.

6.1.

2022 p. 174

2

(1)

(2)

(1)

(2)

6.2.

2025

^{2} $_1$, $_2$, ⋯, .

(1) $_1$, $_2$, ⋯,

(2) .

(3) $= \ _1 + \ _2 + \ +$

.

(4)

(1)

(2) $> 0 : \lim_{\to\infty} (|\bar{\ } - \ | < \ ) = 1$

(3) $\bar{\ } = \frac{}{}$ $(\ , \frac{^2}{})$

$= \bar{\ } \ (\ , \ ^2\frac{^2}{}) = (\ , \ ^2)$

(4)

6.3.

2025 p. 179

1180 20

(1) 25 1170

(2) 0.9 1175

(1) 25 $_1, \ _2, \cdots, \ _{25}$ 1180 20^2

$$\bar{\ } \quad (1180, \frac{20^2}{25})$$

$$(\bar{\ } > 1170) = (> \frac{1170 - 1180}{4}) = (> -2.5) = 0.9938$$

(2) $\bar{\ } \quad (1180, \frac{20^2}{\ })$

$$(\bar{\ } > 1175) = (> \frac{1175 - 1180}{20/\sqrt{\ }}) = (> -\frac{\sqrt{\ }}{4})$$

$$(> -1.29) \approx 0.9015 > 0.9$$

$-1.29 \geq -\frac{\sqrt{\ }}{4}$ $\sqrt{\ } \geq 1.29 \times 4 = 5.16$ $\geq (5.16)^2 = 26.6256$

≥ 27

6.4.

2025

1.2 kg 0.1 kg

100 122.5 kg

$_1, \cdots, \ _{100}$ $= \ _1 + \ + \ _{100}$

$[\ ] = 100 \times 1.2 \text{ kg} = 120 \text{ kg}$ $[\ ] = \sqrt{100} \times 0.1 \text{ kg} = 1 \text{ kg}$

$(120 \text{ kg}, 1 \text{ kg})$

$$(< 122.5 \text{ kg}) = (< \frac{122.5 - 120 \text{ kg}}{1 \text{ kg}}) = (< 2.5) = 0.9938$$

6.5.

2022 pp. 202 - 205

2 3 1 2 3

(1) .

(a) $_1 = {}_1 + {}_2 - {}_3$

(b) $_2 = \frac{_1 + 2 {}_2 + {}_3}{4}$

(c) $_3 = \frac{_1 + {}_2 + {}_3}{3}$

(2) $\{ {}_1, {}_2, {}_3 \}$

---

(1)

(a) $({}_1) = ({}_1) + ({}_2) - ({}_3) = + - =$

(b) $({}_2) = \frac{1}{4} ({}_1) + \frac{2}{4} ({}_2) + \frac{1}{4} ({}_3) = \frac{1}{4} + \frac{2}{4} + \frac{1}{4} =$

(c) $({}_3) = \frac{1}{3} ({}_1) + \frac{1}{3} ({}_2) + \frac{1}{3} ({}_3) = \frac{1}{3} + \frac{1}{3} + \frac{1}{3} =$

(2) 3

(a) $({}_1) = 1^2 {}^2 + 1^2 {}^2 + (-1)^2 {}^2 = 3 {}^2$

(b) $({}_2) = (\frac{1}{4})^2 {}^2 + (\frac{2}{4})^2 {}^2 + (\frac{1}{4})^2 {}^2 = \frac{3}{8} {}^2$

(c) $({}_3) = (\frac{1}{3})^2 {}^2 + (\frac{1}{3})^2 {}^2 + (\frac{1}{3})^2 {}^2 = \frac{1}{3} {}^2$

6.6.

2023 2023 2025 pp. 176 - 180, 202 - 205

(1) 0 1.2 3

$\{-1, 0, 1\}$

(a)

(b)

(c)

(2)

(3)

(4) $\Theta = (\ _1, \ _2, \dots, \ )$ 3

(a)

(b)

(c)

(5) 3

(1) (a) 1.2

(b) $\frac{1}{3}[(-1)^2 + 0^2 + 1^2] = \frac{2}{3}$

(c) $\frac{3}{2} \times \frac{2}{3} = 1$

(2)

(3)

(4) (a) $\hat{}$

$[\hat{} \ ] =$

(b) $\hat{}$

$> 0 : \lim_{\to \infty} (|\hat{} \ - \ | < \ ) = 1$

(c) $\hat{}$ $\hat{}\ ' : \ (\hat{} \ ) \leq \ (\hat{}\ ')$

(5) $\to \infty$

6.7.

2022 2024 p. 205

1000 10

$$32,\quad 86,\quad 78,\quad 49,\quad 48,\quad 77,\quad 97,\quad 49,\quad 51,\quad 74$$

1000

$$\overline{} = \frac{1}{10}(32 + 86 + 78 + 49 + 48 + 77 + 97 + 49 + 51 + 74) = 64.1$$

$$\overline{^2} = \frac{1}{10}(32^2 + 86^2 + 78^2 + 49^2 + 48^2 + 77^2 + 97^2 + 49^2 + 51^2 + 74^2) = 4502.5$$

$$^2 = \overline{^2} - (\overline{})^2 = 4502.5 - (64.1)^2 = 4502.5 - 4108.81 = 393.69$$

$$^2 = \frac{10}{9}\ ^2 = \frac{10}{9} \times 393.69 \approx 437.43$$

6.8.

2025 pp. 179, 216

(1) .

(2)

(3) $_0$ 95%

(4) $= 0.05$ $= 1900$ $(76 \leq \quad \leq 114)$

(1) $(\ ,\ )$

(2)

$(\quad,\quad(1-\quad))$

(3) $2\ \left(\frac{}{2}\right)\sqrt{\frac{{}_0(1-\ {}_0)}{}}$

$$\left(\frac{}{2}\right)\sqrt{\frac{{}_0(1-\ {}_0)}{'}}=\frac{1}{2}\ \left(\frac{}{2}\right)\sqrt{\frac{{}_0(1-\ {}_0)}{}}$$

$$'\qquad '=4$$

(4) $=1900\quad 0.05=95\qquad(1-\quad)=1900\quad 0.05\quad 0.95=90.25=9.5^2$

$(95,9.5^2)$

$$(76\le\quad\le 114)\approx\quad(75.5<\quad<114.5)=\quad\left(\frac{75.5-95}{9.5}<\quad<\frac{114.5-95}{9.5}\right)$$

$$\approx\quad(-2.05<\quad<2.05)\approx 0.9596$$

6.9.

2022 pp. 180 - 182

10,000

3,500 $=\frac{3500}{10000}=0.35$

(1)

(2) 2

(1) $(\quad,\quad(1-\quad))$ 95%

$$\left[0.35-1.96\sqrt{\frac{0.35\quad 0.65}{10000}},0.35+1.96\sqrt{\frac{0.35\quad 0.65}{10000}}\right]\approx[0.3407,0.3593]$$

0.35

(2) $10,000\times 2^2=40,000$

6.10.

2024 2024 2025 pp. 224 - 225

(1) $(1,\quad)$ $_1,\quad{}_2,\cdots,$

(a)

(b)

(2) A

(a) = — A 100(1 −)%

(b)

(3) S TV T

140 32 T

(a) 95%

(b) 3% 0.95

(c)

(1) (a) $[\ ] = \ ,\quad \mathrm{Var}(\ ) = \frac{(1-\ )}{\ }$

(b) $= \ _1 + \ _2 + \ + \quad (\ ,\ (1-\ ))$

$$[\ ] = \ ,\quad \mathrm{Var}(\ ) = \ (1-\ )$$

$$[\ ] = \ [-] = \frac{1}{\ }\ [\ ] = \text{—} = \ ,$$

$$\mathrm{Var}(\ ) = \mathrm{Var}(-) = \frac{1}{\ ^2}\mathrm{Var}(\ ) = \frac{(1-\ )}{\ ^2} = \frac{(1-\ )}{\ }$$

(2) (a) $$[\ -\ (\frac{\ }{2})\sqrt{\frac{(1-\ )}{\ }},\ \ +\ (\frac{\ }{2})\sqrt{\frac{(1-\ )}{\ }}]$$

(b)

(3) (a) $$[\frac{32}{140} \pm 1.96\sqrt{\frac{32 \times (140-32)}{140^3}}] \approx [0.2286 \pm 0.06956] \approx [0.1590, 0.2982]$$

(b) ≈ 0.2286

$$|\ -\ | = 1.96\sqrt{\frac{0.2286 \times 0.7714}{\ }} \leq 0.03$$

$$\geq \frac{0.2286 \times 0.7714 \times 1.96^2}{0.03^2} \approx 752.706$$

753

(c)

$$2 \times 1.96 \sqrt{\frac{(1 - \ )}{}} \leq 2 \times 0.03$$

$$(1 - \ ) \leq \frac{+ (1 - \ )}{2}^2 = \frac{1}{4}$$

AM ≥ GM

$$\geq \left(\frac{1.96}{0.03}\right)^2 \times \frac{1}{4} \approx 1067.1$$

1068

6.11.

2024 pp. 223 - 225

(1) 0.8 0.9 0.99

ˆ 0.02

(2) 0.95 ˆ

0.03

(1)

$$= \frac{\hat{\ } -}{\sqrt{(1- \ )/}}$$

$(0, 1)$ $(| \ | < 2.576) \approx 0.99$

$$(|\hat{\ } - \ | < 2.576 \sqrt{\frac{(1 - \ )}{}}) = 0.99$$

$|\hat{\ } - \ | < 0.02$

$$2.576\sqrt{\frac{(1-\ )}{}} \leq 0.02 \qquad \geq \left(\frac{2.576}{0.02}\right)^2 (1-\ )$$

$[0.8, 0.9]$

$$(1-\ ) \leq 0.8 \times 0.2 = 0.16$$

$$\geq \left(\frac{2.576}{0.02}\right)^2 \times 0.16 \approx 2654.3$$

2655

(2)

$$(1-\ ) \leq \frac{1}{4}$$

$$\geq \left(\frac{1.96}{0.03}\right)^2 \times \frac{1}{4} \approx 1067.1$$

1068

6.12.

2025

2 $1-$

$(\ ;\ )$

(1) $(\ ;\ )$ $(\ )$.

(2) $(\ ;\ )$.

(3) $(\ )$.

(4) .

(5) . .

(a) (4) ?

(b) (4) ?

(c) ~~(4) ?~~ $\longleftarrow$

(1)

$$(\ ) = (\quad)\quad(1 - \ )^{\ -\ }$$

(2)

$$[\ ] = \sum_{=0}\ (\ )$$

$$= \sum_{=0}\ (\quad)\ (1 - \ )^{\ -\ }$$

$$= \sum_{=0}\ \frac{!}{!(\ -\ )!}\ (1 - \ )^{\ -\ }$$

$$= \sum_{=0} \frac{(\ -1)!}{(\ -1)!\{(\ -1) - (\ -1)\}!}\ ^{-1}(1 - \ )^{(\ -1)-(\ -1)}$$

$$= \sum_{'=0}^{-1} \frac{(\ -1)!}{'!\{(\ -1) - \ '\}!}\ ^{'}(1 - \ )^{(\ -1)-\ '}$$

$$= \sum_{'=0}^{-1} \binom{\ -1}{\ '}\ ^{'}(1 - \ )^{(\ -1)-\ '}$$

$$=$$

(3)

$$\prod_{=1}^{1}$$

$$(\ ) = \log\ (\ ;\ ) = \log(\quad)\ (1 - \ )^{\ -\ } = \log(\quad) + \ \log\ + (\ -\ )\log(1 - \ )$$

(4)

$$\frac{}{}(\ ) = \frac{}{} - \frac{-}{1 -} = 0$$

$$\hat{\ } = \frac{}{}$$

(5) (a)

$$[\hat{\ }] = \ [\frac{}{}] = \frac{1}{}\ [\ ] = \frac{1}{}(\quad) =$$

(b)

$$\lim_{\to\infty} \hat{\ } = \lim_{\to\infty} \frac{}{} =$$

(c) **AI**

2026

6.13.

2025

$(\;,\; ^2)$ ^{2}

(1)

(2) ^{2}

(1) $(\;,\; ^2)$ 1 $_1$ $_1 =$

$$[\;\;] = \qquad _1 \quad \hat{}_1 = \frac{1}{} \sum_{=1} \quad = \overline{\;\;}$$

$$\hat{} = \overline{\;\;}$$

$_1,\ _2,\cdots,$ $(\;,\; ^2)$

$_1,\ _2,\cdots,$

$$(\;;\;) = \frac{1}{\sqrt{2}} \, ^{-\frac{(\;-\;)^2}{2\;\;^2}}$$

$$\begin{aligned}
\overline{\quad} \log \; (\;) &= \sum_{=1} \overline{\quad} \log \; (\;;\;) \\
&= \sum_{=1} \overline{\quad} \log \left[\frac{1}{\sqrt{2}} \, ^{-\frac{(\;-\;)^2}{2\;\;^2}} \right] \\
&= \sum_{=1} \overline{\quad} \left[\log \frac{1}{\sqrt{2}} - \frac{(\;-\;)^2}{2\;\;^2} \right] \\
&= \sum_{=1} \frac{-}{\;^2} \\
&= \frac{\overline{\;}-}{\;^2} = 0
\end{aligned}$$

$$= \overline{\;} \qquad\qquad \hat{} = \overline{\;\;}$$

(2) 1 2 $_1$ $_2$

$$_1 = [\;], \qquad _2 = [\;^2]$$

$$^2 = [\;^2] - [\;]^2 = \;_2 - \;_1^2$$

$$\hat{\ }_1 \qquad \hat{\ }_2 = \frac{1}{\ } \sum_{=1} {}^2 = \overline{\ }^2$$

$$\hat{\ }^2 = \hat{\ }_2 - \hat{\ }_1{}^2 = \overline{\ }^2 - \overline{\ }^2$$

$= \ ^2$

$$(\ ; \ ; \ ^2) = \frac{1}{\sqrt{2}} \ ^{-\frac{(-)^2}{2}}$$

$$\frac{}{} \log \ (\ , \ ) = \sum_{=1} \frac{}{} \log \ (\ ; \ ; \ )$$

$$= \sum_{=1} \frac{}{} \log \left[\frac{1}{\sqrt{2}} \ ^{-\frac{(-)^2}{2}} \right]$$

$$= \sum_{=1} \frac{}{} \left[-\frac{1}{2} \log(2 \ ) - \frac{(\ - \ )^2}{2} \right]$$

$$= \sum_{=1} \left[-\frac{1}{2} + \frac{(\ - \ )^2}{2 \ ^2} \right]$$

$$= -\frac{}{2} + \frac{1}{2 \ ^2} \sum_{=1} (\ - \ )^2 = 0$$

$$\hat{\ } = \overline{\ }$$

$$\hat{\ } = \frac{1}{\ } \sum_{=1} (\ - \ )^2 = 0$$

6.14.

2025

(1)

(2)

/ ^{2}

(3) /

(1)

(2)

$$\mathrm{Var}(\overline{\ }) = \frac{\ ^2}{\ }\frac{-}{-1}$$

$$\frac{-}{-1}$$

(a) $= \frac{}{}$ $\leq 5\%$

10000 5

7.

① 仮説の検定

② 統計量の検定

③ 優位水準と棄却域の設定

④ 帰無仮説の棄却・選択

7.1.

2022 2023 pp. 227 - 231

(1)

(2)

(3) (4)

5% (3)

5% (5)

(1)

(1)

(6) (6) (7)

- (8:)
- (9:)

7.2.

2022 pp. 227 - 234

(1)

(2)

7.3.

2023 pp. 2235 - 236

8000

25 7700

5%

640

7.4.

2023 pp. 238 - 240

T

S 365 6

330, 350, 385, 321, 340, 365

5%

7.5.

2025 pp. 238 - 240

T

S 360 6

350, 355, 385, 390, 370, 365

(1) 10%

(2) 5%

7.6.

2023 2024 2024 2025 pp. 235 - 236

(1) $(\ , 49)$ 49

$_0 : \ = 4,$ $_1 : \ > 4$

(a) 0.05 $= 6$ 2

(b) 0.01 $= 7$ 2

(2) $_1$: $= 5$ 0.05

2 0.2 0.1

7.7.

2022 pp. 241 - 242

(1) 2 A B 2

10 12

A 2.04 0.03 B

2.02 0.02 2

5%

2

(2)

5 1 5%

7.11.

2023 pp. 252 - 253

10

$_0 : \; = \frac{1}{2}$ $_0$

(1)

(2)

(a) $_1 : \; < \frac{1}{2}$ 5%

(b) $_1 : \; \neq \frac{1}{2}$ 2%

7.12.

2023 pp. 252 - 253

5

1

(1) 5 5 0

(a) $= \frac{1}{2}$

(b) $= \frac{2}{3}$

(2) $\geq \frac{1}{2}$

(a) 3 $= \frac{1}{2}$

(b) $= \frac{3}{4}$ 3

7.13. 2

256 116 5%

.

(1) 2

(2) 2 10%

7.14.

2023 2025 pp. 252 - 253

$0 < \quad < 1$

$_0$: $= \frac{1}{2}$ 3 3

3 $_0$ $_0$

(1) 1 1

(2) 2 2

(3) 2 $\frac{1}{3}$

7.15.

2024 pp. 252 - 253

3 A B C A 3 B 2 C 1 1

1

(1) A A $\frac{1}{16}$ B B $\frac{1}{9}$ C C $\frac{1}{25}$

1

(2) 960 640

(1) 5%